Luis Martinez
Electrical Engineer

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PROFESSIONAL EXPERIENCE

SPACE INFORMATION LABORATORIES
Electrical Engineer II **2025-Present**

- Testing
  - Helped design electrical test fixtures and procedures for PCBA DVT
  - Designed system test equipment PCBAs to facilitate streamlined testing
  - Wrote test firmware for Xilinx SoC which assisted PCBA and module level DVT
  - Conducted GNSS receiver PVT accuracy and reacquisition characterization tests
- R&D Proposal Support
  - Assisted with proposal efforts by conducting trade studies and generating preliminary ICDs
- Circuit Analysis
  - Used worst-case circuit analysis (WCCA) to evaluate BMS circuit compliance to PMAP Rev B and RCC-319
  - Created FMECA reports for Autonomous Flight Termination Unit and BMS systems

RAYTHEON INTELLIGENCE AND SPACE
Electrical Engineer II **2024-2025**
Electrical Engineer I **2022-2024**

- PCB Design
  - Designed two boards based on customer requirements which were brought through CDR.
  - Used Hyperlynx SI/PI to run power and signal integrity of routed PCBs
  - Created PWB, PWA, parts list, and schematic drawings
- Circuit Analysis
  - Used worst-case circuit analysis techniques (WCCA) to evaluate the performance of various power electronics circuits
- Firmware Design
  - Implemented and debugged raw HDL for Intel SoCs in VHDL using Signaltap, Quartus Pro, and Qsys
  - Used Simulink and MATLAB for higher level system design
- Lab Experience
  - Conducted system level environmental tests with vibe/temperature chambers
  - Frequent work with signal generators, DC power supplies, spectrum analyzers, and oscilloscopes

BOSTON SCIENTIFIC
Electrical Engineering Intern **2021**

- Designed a PCB to model behavior of the spine under certain stimuli
- Wrote C for a microcontroller on the PCB to define response to input data
- Used concepts of DSP, digital and analog filtering, and digital to analog conversion

PERSONAL PROJECTS

Guitar Headphone Amplifier

- Designed a 2-layer PCB which amplifies an electric guitar signal for use with a pair of in-ear headphones. Leverages the NE5532 op amp for active filtering and amplification.

GNSS and IMU PCBA with STM32 and RF Transceiver

- Designed a PCB centered around the NEO-M9N GNSS receiver, STM32F411 MCU, and BNO086 9DOF IMU. Includes an nRF24L01 2.4GHz RF transceiver with external PA/LNA circuitry

SKILLS

- Xpedition Enterprise
- Circuit Simulation (LTSpice)
- VHDL, Verilog, SystemVerilog
- C/C++
- Matlab/Simulink
- FMECA
- Python
- Altium
- Embedded Systems
- VPX/VITA Standard Knowledge

EDUCATION

CORNELL UNIVERSITY **Ithaca, NY**
Bachelor of Science in Electrical and Computer Engineering **2018-2022**

- Coursework centered around Embedded Systems, Digital Signal Processing, and FPGA development
- Electrical lead for Cornell Nexus which strived to create a robot that filters microplastics from beaches. Worked closely with motors and motor controllers, motor encoders, Raspberry Pi’s, and batteries.